

Scrambled when wrong, left alone when right: Outcome-dependent diffusion and learning in cultured neurons playing Pong

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A system whose parameters receive random kicks of variance σ_m^2 after a failure and σ_h^2 after a success diffuses with a state-dependent coefficient $D \propto \kappa + (1 - \kappa)P$, where P is the local failure probability and $\kappa = \sigma_h^2/\sigma_m^2$. With departure-point (Itô) steps the stationary density is $\rho \propto 1/D$: failures become rarer than chance for $\kappa < 1$, unchanged at $\kappa = 1$, more frequent for $\kappa > 1$. Mapped to an imaginary-time Schrödinger operator, the dynamics has a zero-energy ground state $\phi_0 \propto D^{-1/2}$ and a gap $\propto \sigma_m^2$, so the kick strength sets the learning rate but not the endpoint; the stationary state is a Boltzmann law with potential $\ln D$ at unit temperature, which can be lowered only by kicks that grow with P . We test this against DishBrain, where cortical cultures play Pong and a miss triggers stimulation of twice the amplitude of the hit stimulus plus an 8-s time-out. In the published data only the Stimulus condition improves within a session (+0.152 hits per rally, $p = 0.0002$; Silent −0.03 and No feedback −0.002, not significant) and all conditions end at the same level ($p = 0.22$), as the gap scaling predicts; Rest baselines differ between groups ($p = 7 \times 10^{-5}$). A spiking network calibrated to the cultures’ Rest statistics carries no sensory-to-motor path at any density from 375 to 1.2×10^5 neurons per mm^2 (10^6 neurons): pulses only suppress the motor regions, with a weak up-down bias that would steer the paddle away from the ball. It therefore cannot show closed-loop learning, but its emergent κ under local plasticity is 0.6–0.8 for near-threshold recruitment and above 1 for wide recruitment, set by the time-out rather than the pulse amplitude. An equal time-out after hits and misses decides between the two.

I. INTRODUCTION

Random search that is interrupted by success is among the oldest learning mechanisms: Ashby’s ultrastable homeostat changes its parameters at random until its essential variables stay in range [1], finite automata that switch state only when penalized behave better than chance [2–4], and woodlice gather in damp places by moving more in dry air [5]. In physical terms these are random walks whose step size depends on position, and random walkers accumulate where they move slowly [6–8]. How strongly they accumulate, and in which convention the multiplicative noise must be read, are classical questions of stochastic dynamics [9].

Here the position is the setting of a learning system and the step size depends on its performance. The experimental motivation is DishBrain [10], in which cortical cultures on a high-density electrode array play Pong (Fig. 1). Spikes in four motor regions move a paddle, and the ball’s position is delivered to eight sensory electrodes. After a hit the culture receives a “predictable” stimulus (100 Hz for 100 ms on all eight electrodes at 75 mV); after a miss, an “unpredictable” one (150 mV at 5 Hz at random sites and times for 4 s), followed by 4 s without stimulation before a new serve. Cultures improved within a session, and this was read as evidence that neurons act to make their input predictable, as the free-energy principle proposes [10, 11]. The 150 mV amplitude was chosen, however, to force action potentials “regardless of the state the cell was in, thereby being even more disruptive to the

culture” [10]: the feedback after a miss is less predictable and more disruptive at once, and the published controls do not separate the two. Earlier criticism concerned the terms intelligence and sentience [12, 13]; closer precedents in cultures are the stimulus-regulation principle [14, 15] and learning by stimulation avoidance [16, 17].

We formulate learning by outcome-contingent perturbation as a drift-free diffusion with outcome-dependent diffusivity (Sec. II), solve its stationary state and relaxation exactly (Sec. III), and test the predictions against the published data (Sec. IV) and a spiking network calibrated to the cultures (Sec. V).

II. MODEL

Let $\psi \in \Omega$ be the setting of the system, all variables that determine its play, and $P(\psi) \in (0, 1)$ its probability of missing a ball. Time is counted in ball approaches. After a miss the setting jumps with a symmetric kernel, $K(\psi, \psi') = K(\psi', \psi)$; after a hit it stays. On a finite Ω the distribution evolves as

$$\rho_{t+1} = \rho_t - (I - K) \text{diag}(P) \rho_t. \quad (1)$$

In general both outcomes perturb, with variances σ_m^2 after a miss and σ_h^2 after a hit. For small isotropic steps whose size is set at the point of departure, the setting diffuses with

$$D(\psi) = \frac{1}{2} [\sigma_m^2 P(\psi) + \sigma_h^2 (1 - P(\psi))] = \frac{1}{2} \sigma_m^2 [\kappa + (1 - \kappa)P], \quad (2)$$

$\kappa = \sigma_h^2/\sigma_m^2$, and the density obeys the Itô form of the Fokker–Planck equation with no drift [6, 7, 9],

$$\partial_t \rho = \nabla^2 (D \rho), \quad (3)$$

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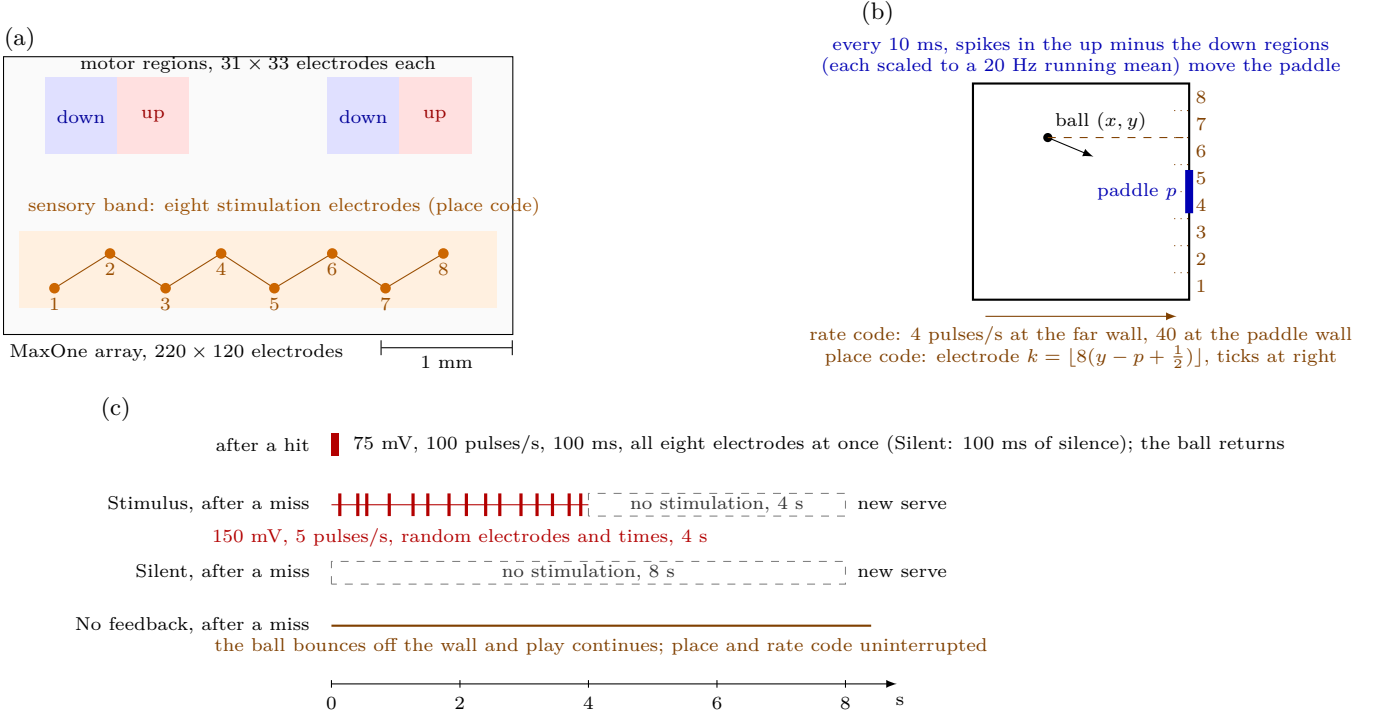


FIG. 1. DishBrain as simulated. (a) The array to scale, with the eight stimulation electrodes of the sensory band in the zig-zag of Fig. 4D of Ref. [10] and the four motor regions, counterbalanced up and down, about 1 mm away; electrode pitch $17.5 \mu\text{m}$. (b) The game: the ball's height relative to the paddle selects the electrode (place code) and its distance from the paddle's wall sets the pulse rate (rate code); every 10 ms the spike counts of the up and down regions, each normalized by a running 20 Hz gain, move the paddle. (c) Feedback after each outcome in the three conditions. Rest sessions run the game with no stimulation at all. Voltages, rates, durations and the electrode layout are from Ref. [10]; court size, paddle half-length and ball speed are the model's choices (Appendix B).

with reflecting boundaries. The Itô convention is not a choice here: a kick is drawn with the variance of the outcome that has just occurred, at the current setting. Nothing in the model computes a prediction, a reward, or the sign of an error.

III. ANALYTICAL RESULTS

A. Stationary state and the sign rule

For the chain (1), $\rho \propto 1/P$ satisfies detailed balance, $\rho(\psi)P(\psi)K(\psi, \psi') = \rho(\psi')P(\psi')K(\psi', \psi)$, and is unique when K is irreducible [18]. The long-run miss rate is then the harmonic mean of P ,

$$\langle P \rangle_\infty = \frac{|\Omega|}{\sum_\psi 1/P(\psi)} \leq \frac{1}{|\Omega|} \sum_\psi P(\psi), \quad (4)$$

below the arithmetic mean, the miss rate of a setting drawn at random, unless P is constant. A setting that misses ten times less often is kept ten times longer.

For the continuum (3) the stationary flux $-\nabla(D\rho)$

vanishes, so

$$\rho_\infty \propto \frac{1}{D} \propto \frac{1}{\kappa + (1 - \kappa)P}. \quad (5)$$

With $w = 1/[\kappa + (1 - \kappa)P]$ and $\mathbb{E}_u$ the uniform average (the stationary law of information-free kicks), $\langle P \rangle_\infty = \mathbb{E}_u[Pw]/\mathbb{E}_u[w]$. For $\kappa < 1$, w decreases with P and Chebyshev's association inequality [19] gives $\langle P \rangle_\infty < \mathbb{E}_u[P]$: learning. At $\kappa = 1$ there is no learning, and for $\kappa > 1$ the inequality reverses: anti-learning. Predictability does not enter; only how far each outcome moves the setting does.

B. Imaginary-time Schrödinger form

The substitution $\rho = D^{-1/2}\phi$ turns Eq. (3) into

$$-\partial_t \phi = H\phi, \quad H = -\sqrt{D} \nabla^2 \sqrt{D}, \quad (6)$$

with the Neumann condition $\partial_n(\sqrt{D}\phi) = 0$ at reflecting walls [20]. Green's identity gives $\langle \chi, H\phi \rangle = \int \nabla(\sqrt{D}\chi) \cdot \nabla(\sqrt{D}\phi)$, so H is self-adjoint and $H \geq 0$. Its ground state is $\phi_0 \propto D^{-1/2}$ with $E_0 = 0$, and $\rho_\infty = \hat{\phi}_0^2$. In one dimension, $dy = d\psi/\sqrt{D}$ and $\chi = D^{1/4}\phi$ bring H to the

standard form $-\partial_y^2 + V(y)$ with

$$V = \frac{3}{4} \left(\frac{q'}{q} \right)^2 - \frac{1}{2} \frac{q''}{q} = \frac{\chi_0''}{\chi_0}, \quad q = \sqrt{D}, \quad \chi_0 = D^{-1/4}, \quad (7)$$

primes denoting d/dy . On a finite set, $\phi = \text{diag}(\sqrt{P})\rho$ turns Eq. (1) into $\phi_{t+1} = \phi_t - H\phi_t$ with the tight-binding Hamiltonian

$$H = \text{diag}(\sqrt{P}) (I - K) \text{diag}(\sqrt{P}), \quad (8)$$

whose ground state is $\phi_0 \propto P^{-1/2}$ with $H\phi_0 = 0$, which reproduces Eq. (4). The evolution is in imaginary time and reversible; nothing is quantum.

The first excited level sets the rate of learning, $\|\rho(t) - \rho_\infty\| \sim e^{-E_1 t}$. When a hit perturbs nothing, $\sigma_h = 0$ and $D = \frac{1}{2}\sigma_m^2 P$, so $H = \sigma_m^2 H_1$ with H_1 fixed by P alone: every eigenvalue scales as σ_m^2 , while every eigenfunction, ρ_∞ included, is independent of σ_m . The strength of the post-failure perturbation sets how fast the system learns, not where it ends. Numerically (script `theory_check.py`), on a random 60-state chain $\max |H\phi_0| = 2 \times 10^{-16}$, the stationary law equals $\hat{\phi}_0^2$ to 10^{-15} , and the miss rate is the harmonic mean 0.298 against the arithmetic mean 0.443; on a one-dimensional landscape $E_1/\sigma_m^2 = 2.6426$ for $\sigma_m = 0.5, 1, 2$ with an unchanged endpoint.

C. Boltzmann form and cooling

Equation (3) can be written as gradient dynamics,

$$\partial_t \rho = \nabla \cdot [D(\nabla \rho + \rho \nabla U)], \quad U = \ln D, \quad (9)$$

so that

$$\rho_\infty \propto e^{-U/T}, \quad U = \ln[\kappa + (1 - \kappa)P], \quad T = 1, \quad (10)$$

up to an additive constant in U , an equilibrium with detailed balance in any dimension, with $U = \ln P$ at $\kappa = 0$. There is no force: the potential is induced by the state dependence of the noise, as in Landauer's "blowtorch" [8]. Three consequences follow. (i) The temperature is pinned at 1: rescaling all kicks by c shifts U by $2 \ln c$ and leaves ρ_∞ unchanged, the same speed-endpoint separation as above. (ii) Concentration is weak: occupancies stand in the ratio of the miss rates, not exponentially in them, which is why the gain is only the gap between the arithmetic and harmonic means. (iii) The system can be cooled: if the per-approach variance grows as P^α , then $\rho_\infty \propto P^{-\alpha}$, an effective temperature $1/\alpha$. On the test landscape the stationary miss rate falls from 0.323 ($\alpha = 1$) to 0.210 ($\alpha = 2$) and 0.155 ($\alpha = 3$), against 0.454 for a random setting and 0.109 for the best one. Finally, any Markov process with stationary law e^{-U} lowers $F[\rho] = \langle U \rangle_\rho - S[\rho]$, the relative entropy to ρ_∞ up to a constant. Here U is the logarithm of the failure rate, not a sensory surprise, so a decrease of free energy during learning does not by itself indicate predictive processing.

D. A closed-loop Pong model

A two-parameter paddle, $p = \text{clip}(ay + b)$ for a ball at height $y \sim U(0, 1)$, with a miss moving (a, b) to a random neighbor on a 31×21 grid, has a predicted stationary miss rate of 0.815 and a simulated one of 0.810, against 0.845 for a random setting. In a DishBrain-like loop (Appendix A) the ball is place- and rate-coded on eight channels as in the experiment, two motor populations read it through 16 plastic weights, and the only teacher is a random kick to all weights after each approach. Over 400 simulated cultures the within-session gain in hit fraction is $+0.161 \pm 0.014$ for $(\sigma_h, \sigma_m) = (0, 0.08)$, $+0.068 \pm 0.012$ for a weaker $(0, 0.03)$, $+0.017 \pm 0.007$ without kicks, -0.010 ± 0.007 for equal kicks and -0.019 ± 0.006 for reversed ones $(0.08, 0.02)$; performance falls monotonically with κ and crosses the information-free level at $\kappa = 1$ (Fig. 2). The model is not fitted to data. In the experiment a rally lasts $12.7 + 5.9h$ s with h hits (fit to the gameplay data), so a 20-min session holds about 170 approaches and 95 misses, against 2000 approaches here; a culture learning within one session needs per-miss perturbations about ten times larger.

IV. RE-ANALYSIS OF THE PUBLISHED DATA

The per-rally gameplay data of the feedback study are public [21]. Taking the culture as the unit and hits per rally as the measure, in the first 5 min of gameplay (T1) and the remaining 15 (T2), Stimulus rises from 0.687 to 0.839 ($+0.152$, Wilcoxon $p = 0.0002$, $n = 24$), in every cell line, whereas Silent (-0.033 , $p = 0.35$, $n = 17$) and No feedback (-0.002 , $p = 1$, $n = 15$) do not change. At T2 the three conditions do not differ (Kruskal-Wallis $p = 0.22$): Stimulus reaches, within the session, the level the other groups already had. This is the speed-endpoint separation of Sec. III: the post-miss perturbation sets how fast the level is reached, not the level. Two caveats apply to the published comparisons, which express gameplay relative to each culture's Rest session: the Rest baselines differ between the groups (0.666, 0.775 and 0.787 hits per rally; $p = 7 \times 10^{-5}$), with the Stimulus cultures worst at rest, and the groups differ in cell type (Silent has no mouse cultures). As a suggestive signature of scrambling, the lag-1 correlation of consecutive rally lengths falls from Rest to gameplay in Stimulus ($0.098 \rightarrow 0.026$, $p = 0.0005$) but not in Silent or No feedback; the Stimulus cultures' high Rest persistence carries the same baseline caveat.

V. CALIBRATED SPIKING NETWORK

To ask whether DishBrain's actual stimuli realize $\kappa < 1$ through local plasticity, we simulated 3000 leaky integrate-and-fire neurons on the 3.85×2.10 mm array with the exact protocol: electrode layout, place and rate code, hit and miss programs, four motor regions with the 20 Hz gain normalization, readout blanking, No feedback as one

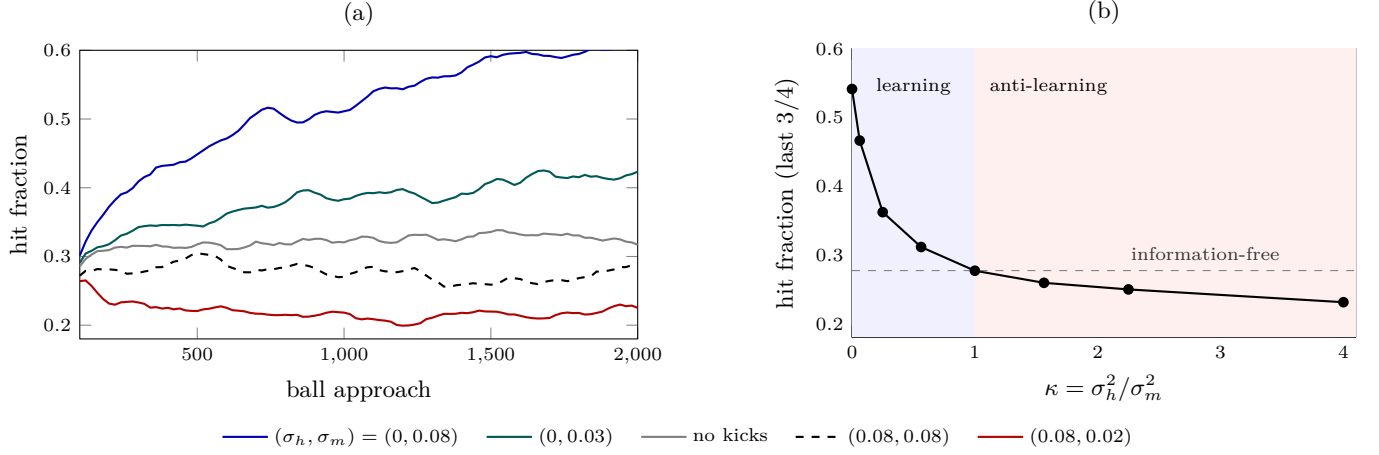


FIG. 2. Learning by outcome-contingent kicks in a closed-loop Pong model (400 simulated cultures). (a) Hit fraction along a session (sliding window of 100 approaches) for five kick schedules. (b) Hit fraction over the last three quarters of a session against κ ($\sigma_m = 0.08$); dashed: the information-free level $\kappa = 1$.

continuous rally, and session timing (Fig. 1; Appendix B). The paddle size gives the published Rest level of 0.73 hits per rally.

Calibration. Free parameters were fitted to eight statistics of the Rest recordings of the human cultures, computed identically for model and data (Table I). The fit lies within the data’s interquartile range except for rates about 0.1 Hz high and pairwise correlations near zero. The Rest statistics fix rates, irregularity, synchrony and the absence of bursts but not recurrent coupling: parameter sets with excitatory weights spanning a factor of 40 fit about equally well, so we carry three (weak, medium and strong coupling).

What the network cannot show. Single pulses at the stimulation electrodes produced no detectable change in the motor regions, about 1 mm away, in any set or recruitment assumption (fraction of cases with $z > 3$: 0.00). In 20-min sessions (medium coupling, activity-gated plasticity, 32 cultures) hits per rally changed by -0.11 , $+0.09$ and $+0.04$ in Stimulus, Silent and No feedback, none significantly ($p = 0.12, 0.07, 0.49$). At 3000 neurons the model has about 375 neurons per mm^2 , far fewer than a culture of 10^6 plated cells, whose responses spread across the array during stimulation [10]. We therefore repeated the test at 10^5 , 3×10^5 and 10^6 neurons (1.2 – 12×10^4 per mm^2) with the same wiring rule, after re-scoring the best parameter sets at each size against the Rest statistics. Density does not remove the coupling degeneracy: at 10^6 neurons the best weak, medium and strong sets have losses 0.71, 1.25 and 1.07. Nor does it create a path. Motor regions were excited in at most 3% of cases at 10^5 and in none at larger sizes. What grows with density is suppression: 150 mV pulses lower motor counts by up to 13%, with $z < -3$ in up to 61, 80 and 97% of cases at the three sizes, alike in up and down regions. The up-minus-down difference exceeds $|z| = 3$ in at most 6% of cases at 10^5 and 3×10^5 and 25% at 10^6 . Its pattern across the eight

electrodes, which are ordered by ball height relative to the paddle, is anticorrelated with correct steering ($r = -0.72$, -0.88 and -0.92 for 150 mV at the three sizes) and similar between sizes although the topologies are independent ($r = 0.47$ – 0.76). The bias is small and consistent, and it pushes the paddle away from the ball; it is not simply proximity to a motor region (a nearest-region prediction correlates with the pattern between -0.45 and 0.64 across sets), and at 10^6 neurons only one culture was run, so agreement across cultures is established at the two smaller sizes. The 500-ms, 40 Hz place-code trains of play give the same result at 10^5 and 3×10^5 neurons: $|z| > 3$ in 3.5 and 6.2% of cases on average (at most 9 and 19%), the sign of an electrode’s effect agrees across cultures in only 22 and 32% of cases, the mean change is 0.2% of the motor counts, and the mean pattern anticorrelates with correct steering ($r = -0.96$ and -0.95 ; $r = 0.92$ between the two sizes, whose topologies are independent). Closed-loop learning therefore cannot be tested in this model at any density we can simulate. What is missing is not cell count but a focused path over about 1 mm, which random distance-dependent wiring with 100 inputs per neuron does not provide.

What it can show: emergent κ . We logged the total squared change of the excitatory weights from each outcome to the next, under two local rules without reward or error signal: activity-gated random updates and spike-timing-dependent plasticity (Table II). When a 75 mV pulse recruits few cells, or many only up to threshold as the authors intended, κ per approach is 0.6–0.8 in all sets and under both rules: the learning regime. It is the same for Stimulus, for a predictable miss stimulus of equal dose, and for Silent, because it comes from time rather than stimulation: a miss is followed by 8 s without play and a new serve, a hit by the return of the ball, and plasticity accrues at a nearly constant rate. The 150 mV random program adds only 15% to the displacement of 8 s of si-

lence with small recruitment. When 75 mV pulses recruit as widely as 150 mV pulses, the 80-pulse hit program dominates and $\kappa > 1$: anti-learning. Under local plasticity at this scale the network therefore predicts learning for Stimulus and Silent alike and none for No feedback, whereas the data show learning in Stimulus only. Either 150 mV stimulation perturbs real synapses far more than activity-gated rules do in this model, or Silent’s missing gain reflects its high starting level (0.849 hits per rally at T1, above Stimulus’s end point).

VI. DISCUSSION

Learning by outcome-contingent perturbation is a noise-induced equilibrium: the potential is the logarithm of the local diffusivity, the temperature is fixed at one, and the only control parameter is κ . The Schrödinger form separates speed from endpoint, which the published DishBrain data show directly: the condition with the strongest post-miss perturbation is the only one that improves within a session, and all conditions end at the same level. The accounts by predictability and by perturbation agree on every published condition. They disagree on the protocols of Table III. B and C change one property of the post-miss feedback while holding the other; D reverses which outcome is more disruptive and tests the sign rule against the information-free level E. F is suggested by the network: equal time-outs after hits and misses, with stimulation only after misses. If the time-out is what perturbs, F gives $\kappa \approx 1$ and no learning; if the stimulation amplitude matters, F learns. Two analyses of existing recordings also decide: a miss should move the network’s state (e.g., the evoked response to a fixed probe) further than a hit, with the within-session gain growing with the difference, and in Silent the number of spontaneous bursts after misses should predict each culture’s gain. Cooling offers a positive test: stimulation that grows with a culture’s recent miss rate should raise the level reached, as $\rho_\infty \propto P^{-\alpha}$.

The limitations are specific. Real stimulation does not perturb synapses isotropically, and a patterned 100 Hz burst can itself induce pathway-specific potentiation and depression [22], so the “predictable” stimulus may push the network in a consistent direction; the displacement analysis measures such effects directly. The network, with random distance-dependent wiring, carries no sensorimotor channel at any density up to 10^6 neurons, so it tests the perturbation contingency, not closed-loop learning; under local rules it attributes $\kappa < 1$ to the time-out rather than to the stimulation and does not explain why Silent did not learn. In bursting cultures, silence after a miss lets bursts return [23], and bursts make weights drift [24]; the human Silent cultures did not burst. Finally, the two accounts need not exclude each other: a surprise-minimizing system may implement exactly this stabilization.

DATA AVAILABILITY

The abstract model (`scrambler.py`), the analytical checks (`theory_check.py`), the GPU network simulator with its calibration and results (`network/`), and the re-analysis of the published data (`reanalysis/`) are available in the folder `papers/dishbrain-scrambler` of <https://github.com/kladkogex/20-watt-computer>; an archived copy will be deposited on Zenodo on publication. The DishBrain data are from Ref. [21].

Appendix A: Closed-loop Pong model

The court is 1×1 with the paddle on the right wall (half-length 0.1). Each serve starts at the far wall with horizontal speed 1 and vertical speed uniform in $[-0.8, 0.8]$; the ball reflects from the other walls. The ball’s height relative to the paddle selects one of eight channels, $k = \lfloor 8(y - p + 0.5) \rfloor$, and its position x sets the rate $s = (4 + 36x)/40$. The motor rates are $[W_{\uparrow, \downarrow}[k] s + \xi]_+$ with Gaussian noise of s.d. 0.35 per 20-ms step, and the paddle moves by $1.5(r_\uparrow - r_\downarrow)$ court heights per second. The 16 weights start uniform in $[-1, 1]$ and reflect at ± 1 , so information-free kicks leave the uniform law invariant. After every approach each weight receives a Gaussian kick of s.d. 0.01, then one of s.d. σ_h after a hit or σ_m after a miss. Each culture plays 2000 approaches; the gain is the hit fraction over the last 1500 minus that over the first 500. In this model a miss is followed by a new serve in all conditions, whereas in the experiment No feedback is one continuous rally; the network of Sec. V plays the exact protocol. The grid check uses $a \in \{-1, -0.9, \dots, 2\}$, $b \in \{-1, -0.9, \dots, 1\}$ and 400 ball heights; moves off the grid or into one of the seven never-miss settings are rejected, which keeps the kernel symmetric, and the chain runs for 3×10^6 approaches.

Appendix B: Network model

Network. $N = 3000$ leaky integrate-and-fire neurons ($\tau_m = 20$ ms, threshold 1, reset 0, refractory period 2 ms) at uniform random positions; 80% excitatory. Each neuron receives 100 inputs drawn without replacement with probability $\propto e^{-d/\lambda}$, $\lambda = 1$ mm, of the order of the input numbers estimated for dissociated cultures [25]; the excitatory fraction and λ are assumptions. The large- N tests keep 100 inputs per neuron and λ , and draw each input’s distance from $d e^{-d/\lambda}$ and its direction uniformly, taking a random neuron in the grid cell at that point, which gives the same law restricted to the sheet. Exponential current synapses ($\tau_e = 5$, $\tau_i = 10$ ms), short-term depression of excitatory synapses (release fraction U , recovery 500 ms), Ornstein–Uhlenbeck membrane noise of stationary s.d. σ , and a fixed per-neuron drive $\mu + \mu_{sd}\xi_i$; integration is exact for the linear parts with $dt = 1$ ms.

TABLE I. Calibration to the Rest recordings of the human cultures of Ref. [10] (data: 168 recordings, 24 chips; model: 64 ten-minute runs of the best set, 81 active units subsampled per run, as in the data). Median [interquartile range].

Statistic	Data	Model
rate, median (Hz)	0.30 [0.26, 0.39]	0.42 [0.38, 0.45]
rate, mean (Hz)	0.55 [0.46, 0.69]	0.72 [0.65, 0.77]
CV of interspike intervals	0.97 [0.93, 1.02]	0.96 [0.96, 0.97]
correlation, 100 ms	0.0025 [0.0007, 0.012]	0.0000 [−0.0002, 0.0001]
Fano factor, population	1.08 [0.92, 1.46]	0.91 [0.89, 0.93]
co-active in 50 ms, 99th pct.	0.086 [0.078, 0.100]	0.086 [0.086, 0.099]
co-active in 50 ms, max.	0.16 [0.13, 0.23]	0.14 [0.12, 0.14]
network events per min	0 [0, 0.15]	0 [0, 0]

TABLE II. Emergent κ in the calibrated network: squared excitatory-weight change from a hit to the next outcome divided by the same after a miss (means over the three coupling sets, 8 cultures each, 5-min sessions). Pred.: predictable 150 mV miss stimulus of equal dose. κ_{prog} : feedback programs alone. Last column: 150 mV miss program relative to 8 s of silence.

Rule	75 mV recruitment	Stim.	Pred.	Silent	κ_{prog}	ratio
gated	small	0.65	0.65	0.63	0.12	1.16
gated	wide, near threshold	0.72	0.72	0.82	0.35	2.71
gated	wide	1.26	1.28	1.11	0.92	2.83
gated	widest	2.95	2.93	1.60	4.60	4.03
STDP	small	0.63	0.63	0.65	0.10	1.15
STDP	wide	1.11	1.10	0.97	0.75	2.55
STDP	widest	2.96	2.94	1.49	4.07	3.21

Calibration. Parameters were drawn at random, 512 sets per round, and refined twice around the best 16 (60, 90 and 120 s of spontaneous activity). The loss is $\sum_s [(v_s - \text{med}_s) / \text{IQR}_s]^2$ over the statistics of Table I, after subsampling the model’s active units to the data’s median of 81. Best set (weak coupling): $\mu = 0.20$, $\sigma = 0.26$, $\mu_{sd} = 0.14$, $J_{EE} = 0.0066$, $J_{IE} = 0.079$, $J_{EI} = 0.066$, $U = 0.089$; medium: $J_{EE} = 0.118$, $J_{EI} = 0.61$; strong:

$J_{EE} = 0.153$, $J_{EI} = 0.47$ (losses 0.52, 0.68, 0.77).

Stimulation and protocol. Electrodes follow Fig. 4D of Ref. [10]: eight stimulation electrodes in a zigzag across the sensory band and four motor regions, counterbalanced up and down. A 75 mV pulse raises the membrane of neurons within r_{75} by k_{75} ; a 150 mV pulse forces a spike within r_{150} . Recruitment (r_{75}, k_{75}, r_{150}): small (0.05 mm, 1.2, 0.10 mm), wide near threshold (0.12, 0.7, 0.20), wide (0.12, 1.2, 0.20), widest (0.20, 1.5, 0.23). The decoder sums spikes per region every 10 ms, scales each region by a running 5-s gain toward 20 Hz, and moves the paddle by 0.01 court heights toward the larger of the up and down sums; readout is blanked for 5 ms after every stimulation command. The ball crosses in 2.9 s; the paddle half-length 0.214 gives 0.73 hits per rally at Rest. A 60-s burn-in without plasticity precedes each session.

Plasticity. Activity-gated rule: when an excitatory neuron spikes, each excitatory input weight changes by $\pm \eta w_{\text{max}} x_j$ with random sign, x_j the input’s 20-ms trace. STDP: additive pair rule with $A_+ = \eta w_{\text{max}}$, $A_- = 1.05 A_+$ and 20-ms traces. Weights stay in $[0, w_{\text{max}}]$, $w_{\text{max}} = 2J_{EE}$.

Implementation and checks. All runs are simulated together on one GPU (NVIDIA GB10): state tensors carry a run axis, synapses propagate only from neurons that spiked, the neuron update is one fused kernel, and each 10-ms block is replayed as a CUDA graph; 288 five-minute sessions take 6.5–8 min. Rates, irregularity and correlations agree with a dense NumPy reference of the same equations within sampling error. At most

TABLE III. Protocols that separate the predictability (Pr.) and perturbation (Pe.) accounts.

Protocol	After hit	After miss	Pr.	Pe.
A published	75 mV, 100 Hz, 100 ms	150 mV random, 4 s; 4 s pause	learns	learns
B pre-dictable penalty	as A	150 mV, same dose, fixed sequence	weak or none	learns
C quenched penalty	as A	8 s random low-amplitude stimulation that suppresses bursts	learns	weak or none
D reversed	150 mV fixed, 1 s	75 mV random, 1 s	learns	below E
E info-free	same random	same random	none	none
F equal time-out	8 s pause	150 mV random, 4 s; 4 s pause	learns	learns only if amplitude matters

12R spikes per millisecond are propagated for R parallel runs; in the 288-run batches this cap was exceeded during synchronous stimulation steps, which drops propagation and plasticity of some spikes during the hit program, so the hit displacement and hence κ are, if anything, underestimated. In the large- N tests the cap is raised so that no spike was dropped. The re-calibration scores the best eight sets of each coupling range from the 3000-

neuron search over 120 s at each size. Single pulses are compared over 50 ms after vs. before, with 30 trials per electrode and 2, 2 and 1 cultures at 10^5 , 3×10^5 and 10^6 neurons; the place-code trains use 4 cultures and 20 trials per electrode, comparing the up-minus-down count over the 500-ms train with the 500 ms before. A 10^6 -neuron run takes 18 min for the pulse test and 41 min for the re-calibration.

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